Q1 - 2024 (04 Apr Shift 1)

The sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is equal to :

(1) 3133

(2) 931

(3) 6131

(4) 633

Q2 - 2024 (04 Apr Shift 1)

Let $a=1+\frac{{}^{2}C_{2}}{3!}+\frac{{}^{3}C_{2}}{4!}+\frac{{}^{4}C_{2}}{5!}+\ldots$,
$b=1+\frac{{}^{1}C_{0}+{}^{1}C_{1}}{1!}+\frac{{}^{2}C_{0}+{}^{2}C_{1}+{}^{2}C_{2}}{2!}+\frac{{}^{3}C_{0}+{}^{3}C_{1}+{}^{3}C_{2}+{}^{3}C_{3}}{3!}+\ldots$
Then $\frac{2b}{a^{2}}$ is equal to

Q3 - 2024 (04 Apr Shift 2)

If the coefficients of x^{4}, x^{5} and x^{6} in the expansion of $(1+x)^{n}$ are in the arithmetic progression, then the maximum value of n is:

(1) 7

(2) 21

(3) 28

(4) 14

Q4 - 2024 (05 Apr Shift 1)

If the constant term in the expansion of $\left(1+2x-3x^{3}\right)\left(\frac{3}{2}x^{2}-\frac{1}{3x}\right)^{9}$ is p, then 108p is equal to

Q5 - 2024 (05 Apr Shift 2)

If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}, x \neq 0$, is $\alpha \times 2^{8} \times \sqrt[5]{3}$, then 25α is equal to :

(1) 724

(2) 742

(3) 639

(4) 693

Q6 - 2024 (06 Apr Shift 1)

If the second, third and fourth terms in the expansion of $(x+y)^n$ are 135,30 and $\frac{10}{3}$, respectively, then $6\left(n^3+x^2+y\right)$ is equal to ______

Q7 - 2024 (08 Apr Shift 1)

Let $\alpha=\sum_{r=0}^n\left(4r^2+2r+1\right){}^nC_r$ and $\beta=\left(\sum_{r=0}^n \frac{{}^nC_r}{r+1}\right)+\frac{1}{n+1}$. If $140<\frac{2\alpha}{\beta}<281$, then the value of n is ______

Q8 - 2024 (08 Apr Shift 2)

If the term independent of x in the expansion of $\left(\sqrt{\mathrm{a}}x^2+\frac{1}{2x^3}\right)^{10}$ is 105 , then a^2 is equal to :

(1) 2

(2) 4

(3) 6

(4) 9

Q9 - 2024 (09 Apr Shift 1)

The coefficient of x^{70} in $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+\ldots+x^{54}(1+x)^{46}$ is ${}^{99}\mathrm{C}_\mathrm{p}-{}^{46}\mathrm{C}_\mathrm{q}$.

Then a possible value of $p+q$ is :

(1) 55

(2) 83

(3) 61

(4) 68

Q10 - 2024 (09 Apr Shift 1)

The remainder when 428^{2024} is divided by 21 is________

Q11 - 2024 (09 Apr Shift 2)

The sum of the coefficient of $x^{2/3}$ and $x^{-2/5}$ in the binomial expansion of $\left(x^{2/3}+\frac{1}{2}x^{-2/5}\right)^9$ is

(1) $21/4$

(2) $63/16$

(3) $19/4$

(4) $69/16$

Answer Key

Q1 (1)	**Q2** (8)	**Q3** (4)	**Q4** (54)
Q5 (4)	**Q6** (806)	**Q7** (5)	**Q8** (2)
Q9 (2)	**Q10** (1)	**Q11** (1)	

Q1

$$T_{r+1}={}^{15}C_r\left(5^{\frac{1}{3}}\right)^r\left(\frac{1}{2^5}\right)^{15-r}$$
$$={}^{15}C_r 5^{\frac{r}{3}}\cdot 2^{\frac{15-r}{5}}$$
$$R=3\lambda, 15\mu$$
$$\Rightarrow r=0,15$$

2 rational terms

$$\Rightarrow {}^{15}C_0 2^3+{}^{15}C_{15}(5)^5$$
$$=8+3125=3133$$

Q2

$$f(x)=1+\frac{(1+x)}{1!}+\frac{(1+x)^2}{2!}+\frac{(1+x)^3}{3!}+\ldots..$$
$$\frac{e^{(1+x)}}{1+x}=\frac{1}{1+x}+1+\frac{(1+x)}{2!}+\frac{(1+x)^2}{3!}+\frac{(1+x)^2}{4!}$$
$$\text{coef } x^2 \text{ in RHS} : 1+\frac{{}^2C_2}{3}+\frac{{}^3C_2}{4}+\ldots=a$$

coeff. x^2 in L.H.S.

$$e\left(1+x+\tfrac{x^2}{2!}\right)\ldots\left(1-x+\tfrac{x^2}{2!}\ldots\ldots\right)$$

is $e-e+\frac{e}{2!}=a$

$$b=1+\frac{2}{1!}+\frac{2^2}{2!}+\frac{2^3}{3!}+\ldots\ldots=e^2$$
$$\frac{2\,b}{a^2}=8$$

Q3

Coeff. of $x^4 = {}^nC_4$

Coeff. of $x^5 = {}^nC_5$

Coeff. of $x^6 = {}^nC_6$

${}^nC_4, {}^nC_5, {}^nC_6 \ldots . AP$

$2C_5 = {}^nC_4 + {}^nC_6$

$$2 = \frac{{}^nC_4}{{}^nC_5} + \frac{{}^nC_6}{{}^nC_5} \quad \left\{ \frac{{}^nC_r}{{}^nC_r} = \frac{n-r+1}{r} \right\}$$

$$2 = \frac{5}{n-4} + \frac{n-5}{6}$$

$12(n-4) = 30 + n^2 - 9n + 20$

$n^2 - 21n + 98 = 0$

$(n-14)(n-7) = 0$

$n_{\max} = 14 \quad n_{\min} = 7$

Q4

$$\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$$

General term $m\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

$$= {}^9C_r \cdot \frac{3^{9-2r}}{2^{9-r}}(-1)^r \cdot x^{18-3r}$$

Put $r = 6$ to get coeff. of $x^0 = {}^9C_6 \cdot \frac{1}{6^3} \cdot x^0 = \frac{7}{18}x^0$

Put $\mathrm{r} = 7$ to get coeff. of $\mathrm{x}^{-3} = {}^9\mathrm{C}_\mathrm{r} \cdot \frac{3^{-5}}{2^2}(-1)^7 \cdot \mathrm{x}^{-3}$

$$= -{}^9C_7 \cdot \frac{1}{3^5 \cdot 2^2} \cdot x^{-3} = \frac{-1}{27}x^{-3}$$

$$\left(1+2x-3x^3\right)\left(\frac{7}{18}x^0 - \frac{1}{27}x^{-3}\right)$$

$$\frac{7}{18} + \frac{3}{27} = \frac{7}{18} + \frac{1}{9} = \frac{7+2}{18} = \frac{9}{18} = \frac{1}{2}$$

$$\therefore 108 \cdot \frac{1}{2} = 54$$

Q5

$$T_{r+1} = {}^{12}C_r \left(\frac{3^{1/5}}{x}\right)^{12-r} \left(\frac{2x}{5^{1/3}}\right)^r$$

$$T_{r+1} = \frac{{}^{12}C_r (3)^{\frac{12-r}{5}} (2)^r (x)^{2r-12}}{(5)^{r/3}}$$

$r = 6$

$$T_7 = \frac{{}^{12}C_6 (3)^{6/5} (2)^6}{5^2} = \left(\frac{9 \times 11 \times 7}{25}\right) 2^8 \cdot 3^{1/5}$$

$25\alpha = 693$

Q6

${}^nC_1 x^{n-1} y = 135....(i)$

${}^nC_2 x^{n-2} y^2 = 30....(ii)$

${}^nC_3 x^{n-3} y^3 = \frac{10}{3}....(iii)$

By $\dfrac{(i)}{(\text{ii})}$

$$\frac{{}^nC_1}{{}^nC_2}\frac{x}{y} = \frac{9}{2}\ldots(iv)$$

By $\dfrac{(\text{ii})}{(\text{iii})}$

$$\frac{{}^nC_2}{{}^nC_3}\frac{x}{y} = 9....(v)$$

By $\frac{(\text{ iv })}{(v)}$

$\frac{{ }^n C_1{ }^n C_3}{{ }^n C_2{ }^n C_2}=\frac{1}{2}$

$\frac{2 n^2(n-1)(n-2)}{6}=\frac{n(n-1)}{2} \frac{n(n-1)}{2}$

$4 n-8=3 n-3$

$\Rightarrow n=5$

put in (v)

$\frac{x}{y}=9$

$x=9 y$

put in (i)

${ }^5 C_1 x^4\left(\frac{x}{9}\right)=135$

$x^5=27 \times 9$

$\Rightarrow x=3, \quad y=\frac{1}{3}$

$6\left(n^3+x^2+y\right)$

$=6\left(125+9+\frac{1}{3}\right)$

$=806$

Q7

$$\alpha=\sum_{r=0}^{n}\left(4r^2+2r+1\right)\cdot{}^{n}C_r$$

$$\alpha=4\sum_{r=0}^{n}r^2\cdot\frac{n}{r}\cdot{}^{n-1}C_{r-1}+2\sum_{r=0}^{n}r\cdot\frac{n}{r}\cdot{}^{n-1}C_{r-}+\sum_{r=0}^{n}{}^{n}C_r$$

$$+4n\sum_{r=0}^{n}{}^{n-1}C_{r-1}+2n\sum_{r=0}^{n}{}^{n-1}C_{r-1}+\sum_{r=0}^{n}{}^{n}C_r$$

$$\alpha=4n(n-1)\cdot 2^{n-2}+4n\cdot 2^{n-1}+2n\cdot 2^{n-1}+2^n$$

$$\alpha=2^{n-2}[4n(n-1)+8n+4n+4]$$

$$\alpha=2^{n-2}\left[4n^2+8n+4\right]$$

$$\alpha=2n(n+1)^2$$

$$\beta=\sum_{r=0}^{n}\frac{{}^{n}C_r}{r+1}+\frac{1}{n+1}$$

$$=\sum_{r=0}^{n}\frac{C^{n+1}C_{r+1}}{n+1}+\frac{1}{n+1}$$

$$=\frac{1}{n+1}\left(1+{}^{n+1}C_1+\ldots.+{}^{n+1}C_{n+1}\right)$$

$$=\frac{2^{n+1}}{n+1}$$

$$\frac{2\alpha}{\beta}=\frac{2^{n+1}(n+1)^2}{2^{n+1}}\cdot(n+1)=(n+1)^3$$

$$140<(n+1)^3<281$$

$$n=4\Rightarrow(n+1)^3=125$$

$$n=5\Rightarrow(n+1)^3=216$$

$$n=6\Rightarrow(n+1)^3=343$$

$$\therefore n=5$$

Q8

$$\left(\sqrt{\mathrm{a}}\mathrm{x}^2+\frac{1}{2\mathrm{x}^3}\right)^{10}$$

General term $={}^{10}\mathrm{C}_\mathrm{r}\left(\sqrt{\mathrm{a}}\mathrm{x}^2\right)^{10-\mathrm{r}}\left(\frac{1}{2\mathrm{x}^3}\right)^{\mathrm{r}}$

$$20-2r-3r=0$$

$$\mathrm{r}=4$$

$${}^{10}\mathrm{C}_4\mathrm{a}^3\cdot\frac{1}{16}=105$$

$$\mathrm{a}^3=8$$

$$\mathrm{a}^2=4$$

SECTION - B

Q9

$x^2(1+x)^{98}+x^3\left(1+x^{97}\right)+x^4(1+x)^{96}+\ldots\ldots$
$x^{54}(1+x)^{46}$

Coeff. of x^{70} : ${}^{98}C_{68}+{}^{97}C_{67}+{}^{96}C_{66}+\ldots\ldots\ldots$
${}^{47}C_{17}+{}^{46}C_{16}$

$={}^{46}C_{30}+{}^{47}C_{30}+\ldots\ldots\ldots\ldots{}^{98}C_{30}$

$=\left({}^{46}C_{31}+{}^{46}C_{30}\right)+{}^{47}C_{30}+\ldots\ldots\ldots\ldots{}^{98}C_{30}-{}^{46}C_{31}$

$={}^{47}C_{31}+{}^{47}C_{30}+\ldots\ldots\ldots\ldots{}^{98}C_{30}-{}^{46}C_{31}$

$\ldots\ldots$

$={}^{99}C_{31}-{}^{46}C_{31}={}^{99}C_p-{}^{46}C_q$

Possible values of $(p+q)$ are $62, 83, 99, 46$

$\Rightarrow p+q=83$

Q10

$(428)^{2024}=(420+8)^{2024}$

$=(21\times 20+8)^{2024}$

$=21\text{ m}+8^{2024}$

Now $8^{2024}=\left(8^2\right)^{1012}$

$=(64)^{1012}$

$=(63+1)^{1012}$

$=(21\times 3+1)^{1012}$

$=2\ln+1$

$\Rightarrow$ Remainder is 1.

Q11

$T_{r+1}={}^9C_r\left(x^{2/3}\right)^{9-r}\left(\frac{x^{-2/5}}{2}\right)^r$

$={}^9C_r\left(\frac{1}{2}\right)^r(r)^{\left(6\frac{2r}{3}\frac{2r}{5}\right)}$

for coefficient of $x^{2/3}$, put $6-\frac{2r}{3}-\frac{2r}{5}=\frac{2}{3}$

$\Rightarrow r=5$

$\therefore$ Coefficient of $x^{2/3}$ is $={}^9C_5\left(\frac{1}{5}\right)^5$

For coefficient of $x^{-2/5}$, put $6-\frac{2r}{3}-\frac{2r}{5}=-\frac{2}{5}$

$\Rightarrow r=6$

Coefficient of $x^{-2/5}$ is ${}^{9}C_{6}\left(\frac{1}{2}\right)^{6}$

Sum $= {}^{9}C_{5}\left(\frac{1}{2}\right)^{5} + {}^{9}C_{6}\left(\frac{1}{2}\right)^{6} = \frac{21}{4}$